Assignment -22

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Lab: 22.2

Task 1 – Bias Detection in AI Suggestions (Alternative Example)

Example Scenario:
Ask AI to generate a resume shortlisting algorithm multiple times.
• Change prompt wording (e.g., “select best candidate” vs “ensure diversity”).
• Compare outputs and logic.
Discussion Points:
• Does the AI favor certain qualifications, backgrounds, or experiences?
• How could biased shortlisting algorithms affect equal employment opportunities?

Prompt:

Generate a Python algorithm to shortlist candidates for a job based on qualifications, experience, and skills. Then modify the algorithm to ensure diversity and fairness in selection. Provide both versions for comparison.

Code:

```
candidates = [
    {"name": "A", "experience": 5, "degree": "BTech"},
    {"name": "B", "experience": 2, "degree": "Diploma"},
    {"name": "C", "experience": 4, "degree": "BTech"}
]

# Biased selection
selected = [c for c in candidates if c["degree"] == "BTech"]

print("Biased Selection:", selected)
```

```
# Fair selection
fair_selected = [c for c in candidates if c["experience"] >= 3]

print("Fair Selection:", fair_selected)
```

Output:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Biased Selection: [{'name': 'A', 'experience': 5, 'degree': 'BTech'}, {'name': 'C', 'experience': 4, 'degree': 'BTech'}]
Fair Selection: [{'name': 'A', 'experience': 5, 'degree': 'BTech'}, {'name': 'C', 'experience': 4, 'degree': 'BTech'}]
PS C:\Users\Aishwarya Nemalipuri>
```

Task 2 – Privacy and Sensitive Data Handling (Alternative Example)

Example Scenario:
Ask AI to write code for processing student academic records (roll number, marks, email).
Discussion Points:
• How should student identity data be masked or encrypted?
• What privacy laws (e.g., FERPA, GDPR-like principles) are relevant?
• What ethical responsibility does the programmer have when using AI-generated code?

Prompt: Write a Python program to process student records including roll number, marks, and email. Then improve it by masking sensitive data and using basic encryption techniques. Explain privacy considerations.

Code:

```
import hashlib

student = {
    "roll": 101,
    "marks": 90,
    "email": "student@gmail.com"
```

```
}

# Mask email
masked_email = student["email"][:3] + "***@gmail.com"

# Encrypt roll number
encrypted_roll = hashlib.sha256(str(student["roll"]).encode()).hexdigest()

print("Roll:", encrypted_roll)
print("Marks:", student["marks"])
print("Email:", masked_email)
```

Output:

```
PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Roll: 16dc368a89b428b2485484313ba67a3912ca03f2b2b42429174a4f8b3dc84e44
Marks: 90
Email: stu***@gmail.com
PS C:\Users\Aishwarya Nemalipuri>
```

Task 3 – AI-Generated Security Risks (Alternative Example)

Example Scenario:

Use AI to generate a payment processing script for an online shopping system.

Student Tasks:

• Identify risks such as:

o Plain-text card data storage

o Missing input validation

• Propose ethical coding standards to prevent financial fraud and Misuse.

Prompt: Generate a Python script for processing online payments. Then identify security risks such as storing card details in plain text and missing input validation. Improve the code with secure practices.

Code:

```
card_number = input("Enter card number: ")
amount = input("Enter amount: ")

# Mask card number
masked_card = "**** **** **** " + card_number[-4:]
```

```
# Validate amount
if not amount.isdigit():
    print("❌ Invalid amount")
else:
    print("Processing payment with card:", masked_card)
```

Output:

```
PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Enter card number: 214785369145
Enter amount: 50000000
Processing payment with card: **** **** **** 9145
PS C:\Users\Aishwarya Nemalipuri>
```

Task 4 – Accountability in Human–AI Collaboration (Alternative Example)

Scenario:

AI generates code for an autonomous traffic signal control system.

Discussion Questions:

• If a software error causes accidents, who is accountable?

• Should responsibility lie with:

o AI tool developers

o Software engineers

o Government authorities deploying the system?

Students submit a short reflection on accountability.

Prompt: Generate a basic code example for an automated traffic signal control system. Then discuss potential risks and explain who is accountable if failures occur, including developers, AI tools, and authorities.

Code:

```
import time
traffic = ["RED", "GREEN", "YELLOW"]
for signal in traffic:
    print("Signal:", signal)
    time.sleep(2)
```

Output:

```
PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Signal: RED
Signal: GREEN
Signal: YELLOW
PS C:\Users\Aishwarya Nemalipuri>
```

Task 5 – Limitations of AI in Critical Systems (Alternative Example)

Example Scenario:
Ask AI to generate code for an aircraft maintenance fault detection system.
Discussion Points:
• Can AI-generated code be trusted without expert validation?
• What human oversight is essential in aviation software?
• How should safety-critical code be reviewed and certified?

Prompt: Generate a simple Python program for detecting faults in an aircraft maintenance system. Then analyze its limitations and explain why human validation and safety checks are required.

Code:

```
try:
    temperature = int(input("Enter engine temperature: "))
    if temperature > 100:
        print("⚠ Fault detected!")
    else:
        print("☑ Normal")
except:
    print("❌ Invalid input")
```

Output:

```
PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Enter engine temperature: 30
✅ Normal
PS C:\Users\Aishwarya Nemalipuri>
```

Task 6 – Ethical Use of AI-Generated Code (Alternative Example)

Example Scenario:

Students analyze cases where AI-generated code is used in academic assignments or startups.
Discussion Points:
• Risks of submitting AI code without understanding it
• Copyright and license violations
Best practices:
o Proper citation
o Manual verification
o Ethical reuse of AI outputs
Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

Prompt: Provide examples of using AI-generated code in academic projects or startups. Discuss ethical concerns such as plagiarism, lack of understanding, and copyright issues, and suggest best practices.

Code:

```
# Written with understanding
def greet(name):
    return f"Hello {name}"
print(greet("Student"))
```

Output:

```
PS C:\Users\Aishwarya Nemalipuri> python lab22.py
Hello Student
PS C:\Users\Aishwarya Nemalipuri>
```